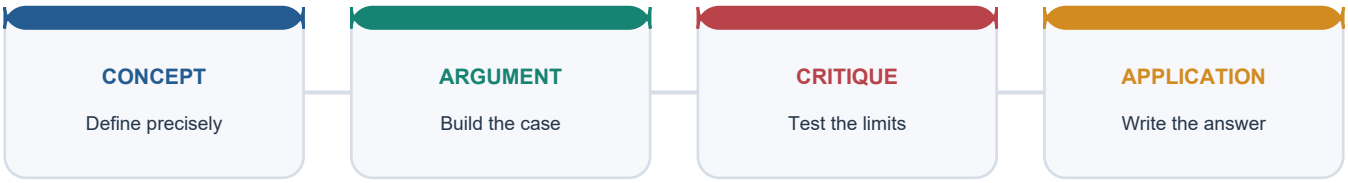
SOLVED PRACTICE WORKBOOK

Earthquake Risk and Resilient Construction - Solved Practice Workbook

GS-III Disaster Management | 32 solved MCQs, PYQ and Mains models

LEARNING PATH



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Earthquake Risk and Resilient Construction - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. A city in Zone III has dense unsafe masonry. Which conclusion is strongest?

A. Risk may be high because vulnerability and exposure matter alongside zone B. Zone III guarantees low loss C. Only Zone V requires codes D. The next earthquake can now be predicted

Answer: A.

Explanation (option-by-option):

- **A.** Correct: zone is one risk component.
- **B.** No zone guarantees an outcome.
- **C.** Codes apply beyond Zone V.
- **D.** Zoning is not prediction.

Trap: Do not turn a regional hazard class into a building-safety certificate.

Q2. A single event damages one town more than another despite the same magnitude. Why?

A. The event had two magnitudes B. Intensity differed because of site and building conditions C. MSK measures plate speed D. NCS changed the energy release

Answer: B.

Explanation (option-by-option):

- **A.** One event has one source magnitude estimate.
- **B.** Correct: intensity varies spatially.
- **C.** MSK is an intensity scale.
- **D.** Reporting does not alter the event.

Trap: Magnitude is not a direct damage score.

Q3. Which cluster best demonstrates that India's seismic risk extends beyond the Himalayan belt?

A. Only the Indo-Gangetic plain B. Only the North-East C. Gulf of Khambhat/Rann of Kutch, peninsular pockets, Lakshadweep and A&N D. Only Zone V districts

Answer: C.

Explanation (option-by-option):

- **A.** The plain is only one part of the risk geography.
- **B.** The North-East is highly exposed but not the whole picture.
- **C.** Correct: the cluster spans western, peninsular and island settings.
- **D.** Risk is not confined to Zone V.

Trap: Use the wider active-zone list to avoid a Himalaya-only answer.

Q4. A planner uses microzonation to announce an earthquake date. What is wrong?

A. Nothing B. Microzonation measures magnitude C. It replaces IS 1893 D. It maps local risk variation, not event timing

Answer: D.

Explanation (option-by-option):

- **A.** Prediction is not available.
- **B.** It includes site effects, not event magnitude.
- **C.** It refines rather than abolishes broader zoning.
- **D.** Correct.

Trap: Local risk mapping is not forecasting.

Q5. A village with heavy unreinforced masonry in peninsular India illustrates:

A. Constructed vulnerability outside the Himalayan stereotype B. Proof the Peninsula is aseismic C. A non-structural issue only D. NCS network failure

Answer: A.

Explanation (option-by-option):

- **A.** Correct: unsafe stock drives risk.
- **B.** Latur and Koyna refute this.
- **C.** The main load-bearing system is structural.
- **D.** Monitoring does not create the vulnerability.

Trap: Use peninsular cases to correct geographical overgeneralisation.

Q6. A hospital frame survives but oxygen lines and generators fail. This is:

A. Complete lifeline resilience B. Non-structural and functional failure despite frame survival C. A prediction failure D. Only a land-use issue

Answer: B.

Explanation (option-by-option):

- **A.** Service continuity failed.
- **B.** Correct.
- **C.** No forecast claim is involved.
- **D.** Siting does not explain equipment failure alone.

Trap: Lifeline safety includes systems and services.

Q7. Which building feature most directly supports ductile behaviour?

A. Heavy unconnected parapets B. Open ground floor without design C. Proper reinforcement detailing and continuous load paths D. Unanchored roof tanks

Answer: C.

Explanation (option-by-option):

- **A.** This adds falling risk.
- **B.** This can create soft-storey vulnerability.
- **C.** Correct.
- **D.** This is non-structural risk.

Trap: Ductility comes from engineered detailing, not a label.

Q8. A building has approved drawings but reinforcement was altered on site. The correct verdict is:

A. Code approval proves safety B. Only maintenance matters C. Zoning is wrong D. As-built verification is missing, so compliance is unproven

Answer: D.

Explanation (option-by-option):

- **A.** Approval is only one link.
- **B.** Execution is the immediate gap.
- **C.** The scenario does not challenge zoning.
- **D.** Correct.

Trap: Design compliance and construction quality are separate evidence stages.

Q9. Which action addresses existing vulnerable stock?

A. Engineering audit followed by designed retrofitting B. Only stronger rules for future buildings C. Printing awareness posters D. Changing the event magnitude

Answer: A.

Explanation (option-by-option):

- **A.** Correct.
- **B.** New-build rules do not fix existing stock.
- **C.** Awareness cannot strengthen a deficient frame.
- **D.** Magnitude cannot be changed.

Trap: Retrofitting follows assessment; it is not generic repair.

Q10. Why should an EOC be a priority lifeline?

A. It is always the tallest building B. Its failure can disable coordination during response C. It issues seismic-zone standards D. It replaces hospitals

Answer: B.

Explanation (option-by-option):

- **A.** Height alone is not the reason.
- **B.** Correct.
- **C.** BIS and planning systems set standards.
- **D.** Functions are different.

Trap: Criticality depends on cascading service consequences.

Q11. A State publishes a hospital retrofit list but allocates no funds. What is established?

A. Completed mitigation B. Functional continuity C. Identification of priority, not execution D. Lower casualties

Answer: C.

Explanation (option-by-option):

- **A.** No work is completed.
- **B.** Systems remain untested.
- **C.** Correct.
- **D.** Outcome cannot be inferred.

Trap: List, sanction, work and verified completion are distinct.

Q12. Which chronology is correct?

A. 2014 six-pillar guideline -> 2019 NBC guideline -> 2021 retrofit guideline B. 2007 retrofit guideline -> 2014 zone map -> 2021 prediction manual C. 2021 six-pillar guideline -> 2014 BhooKamp guide -> 2007 NBC D. 2007 earthquake guideline -> June 2014 retrofit guideline -> May 2021 simplified NBC safety guideline

Answer: D.

Explanation (option-by-option):

- **A.** The functions and years are misplaced.
- **B.** The 2007 document supplies the wider framework, not this sequence.
- **C.** This reverses and mislabels the documents.
- **D.** Correct.

Trap: Keep framework, retrofit and simplified-building guidelines distinct.

Q13. Which statement correctly assigns State responsibility?

A. States identify priority structures such as Raj Bhavans, legislatures, courts, academic institutions, reservoirs, dams and five-plus-storey buildings B. NCS selects every building for retrofit C. BMTPC commands post-impact rescue D. Only Parliament can order local structural audits

Answer: A.

Explanation (option-by-option):

- **A.** Correct.
- **B.** NCS monitors seismicity.
- **C.** BMTPC is a technical promotion body.
- **D.** State and local implementation does not require this exclusive route.

Trap: Name the responsible level and the exact priority categories.

Q14. The Vulnerability Atlas is best used to:

A. Certify one building safe B. Support broad hazard and housing-stock screening C. Replace structural audit D. Prove the latest retrofit completion

Answer: B.

Explanation (option-by-option):

- **A.** Building safety needs specific audit.
- **B.** Correct.
- **C.** Atlas and audit serve different scales.
- **D.** It is not a completion database.

Trap: An atlas is a planning input, not a certificate.

Q15. Which statement about NERMP is safest?

A. Its current budget proves nationwide retrofit completion B. It predicts earthquakes C. It is a named structural/non-structural mitigation route; current status needs dated evidence D. It replaces State responsibility

Answer: C.

Explanation (option-by-option):

- **A.** No such verified outcome is provided.
- **B.** Prediction is impossible.
- **C.** Correct.
- **D.** States retain responsibilities.

Trap: Do not invent current finance or completion data.

Q16. A revised zonation proposal was announced but withdrawn in March 2026. What remains operative?

A. The announced revision B. A five-zone system C. No seismic standard D. IS 1893 (Part 1): 2016 four-zone scheme

Answer: D.

Explanation (option-by-option):

- **A.** Withdrawal prevents operation.
- **B.** The operative system has four zones.
- **C.** Standards remain.
- **D.** Correct.

Trap: Announcement is not implementation.

Q17. Which set consists only of earthquake-preparedness support tools?

A. Homeowner safety manuals, structural-audit manuals, vulnerability maps and organised volunteer support B. Moment magnitude, plate speed, fault depth and zone area C. Only rescue equipment purchased after impact D. A retrofit list with no audit or plan

Answer: A.

Explanation (option-by-option):

- **A.** Correct.
- **B.** These are hazard descriptors, not preparedness tools.
- **C.** Preparedness is wider and begins before impact.
- **D.** A list alone is not a preparedness system.

Trap: Separate preparedness tools from physical hazard measurements.

Q18. Why can licensing of engineers and masons matter?

A. It guarantees zero damage B. It supports competence and accountability in the compliance chain C. It predicts shaking D. It changes zone boundaries

Answer: B.

Explanation (option-by-option):

- **A.** No system guarantees zero damage.
- **B.** Correct.
- **C.** Licensing is not prediction.
- **D.** Zones are hazard classifications.

Trap: Professional licensing is one link, not the whole chain.

Q19. A town allows dense construction on liquefaction-prone ground because buildings meet generic code. What is missing?

A. Only a drill B. NCS app access C. Risk-sensitive siting and local site requirements D. A higher magnitude

Answer: C.

Explanation (option-by-option):

- **A.** Drills do not resolve siting.
- **B.** The app does not change ground failure.
- **C.** Correct.
- **D.** Magnitude is not a planning choice.

Trap: Codes and land-use controls must respond to site conditions.

Q20. Which governance statement is correct?

A. Only the Centre identifies every priority building B. Only communities enforce building codes C. Private builders set public hazard zones D. States identify priority structures while all levels ensure safety within their control

Answer: D.

Explanation (option-by-option):

- **A.** State-specific identification matters.
- **B.** Communities cannot replace regulators.
- **C.** Public standards are not private choices.
- **D.** Correct.

Trap: Avoid vague 'government' attribution.

Q21. NCS's 170-observatory network primarily supports:

A. Detection, analysis and reporting of seismic events B. Exact future prediction C. Building approval D. Retrofitting finance

Answer: A.

Explanation (option-by-option):

- **A.** Correct.
- **B.** Monitoring is not prediction.
- **C.** Local planning systems approve construction.
- **D.** Finance is separate.

Trap: Network scale is monitoring capacity, not warning outcome.

Q22. What is BhooKamp?

A. A new seismic zone B. NCS's official earthquake-information app C. A retrofit code D. An EOC command system

Answer: B.

Explanation (option-by-option):

- **A.** It is not zoning.
- **B.** Correct.
- **C.** Codes are separate.
- **D.** IRS/EOC handles response coordination.

Trap: Use BhooKamp rather than the stale India Quake label.

Q23. NCS rapidly reports an event's parameters after detection. What is the strongest inference?

A. The event was predicted B. The building stock is now safe C. Situational awareness improves, but prior mitigation and preparedness still determine loss D. The seismic zone automatically changes

Answer: C.

Explanation (option-by-option):

- **A.** Detection is not prediction.
- **B.** Reporting cannot strengthen buildings.
- **C.** Correct.
- **D.** One event report does not redraw the operative standard.

Trap: Rapid information is useful without becoming a safety-outcome claim.

Q24. Which current statement is false?

A. IS 1893:2016 remains operative B. The revised zonation was withdrawn C. NCS reports 170 observatories D. India operates precise earthquake prediction

Answer: D.

Explanation (option-by-option):

- **A.** True.
- **B.** True.
- **C.** True.
- **D.** Correct: this is false.

Trap: Prediction overclaim is a recurring trap.

Q25. After shaking, what should EOC activation achieve first?

A. Shared situational awareness and organised incident command B. Redraw national zones C. Approve every building D. Calculate future earthquake dates

Answer: A.

Explanation (option-by-option):

- **A.** Correct.
- **B.** Zoning is not an immediate response task.
- **C.** Building approval is not post-impact command.
- **D.** Prediction is unavailable.

Trap: Response begins with information, priorities and coordination.

Q26. A drill occurs annually but no findings change the plan. The missing element is:

A. Magnitude measurement B. After-action review and corrective revision C. Seismic zoning D. Retrofitting design

Answer: B.

Explanation (option-by-option):

- **A.** No event measurement is required.
- **B.** Correct.
- **C.** The issue is preparedness learning.
- **D.** The scenario concerns drill management.

Trap: Drills are valuable when they improve readiness.

Q27. Which household action is non-structural mitigation?

A. Reinforce the building frame B. Rebuild the foundation C. Anchor heavy shelves and secure cylinders D. Prepare a microzonation map

Answer: C.

Explanation (option-by-option):

- **A.** Structural.
- **B.** Structural.
- **C.** Correct.
- **D.** Planning/technical mapping.

Trap: Household actions reduce contents hazards but cannot fix unsafe frames.

Q28. Why should aftershocks be included in response planning?

A. They alter the original magnitude B. They make codes irrelevant C. They occur only in Zone V D. They can endanger rescuers and further damage weakened structures

Answer: D.

Explanation (option-by-option):

- **A.** Magnitude reporting remains separate.
- **B.** Codes remain relevant.
- **C.** Aftershocks are not zone-exclusive.
- **D.** Correct.

Trap: Response safety must account for unstable structures.

Q29. The 2021 PYQ is best answered through:

A. Hazard setting, vulnerable stock/capacity, regional cases and mitigation B. A list of magnitudes only C. Only Himalayan tectonics D. Only post-disaster relief

Answer: A.

Explanation (option-by-option):

- **A.** Correct.
- **B.** Magnitude lists miss vulnerability.
- **C.** India's risk is wider.
- **D.** The demand includes vulnerability and examples.

Trap: Answer the risk system, not just the hazard.

Q30. Which case best counters 'peninsular India is aseismic'?

A. Only Kashmir 2005 B. Latur 1993 C. Only Sikkim 2011 D. Andaman tsunami

Answer: B.

Explanation (option-by-option):

- **A.** Kashmir is Himalayan.
- **B.** Correct.
- **C.** Sikkim is Himalayan.
- **D.** A tsunami is not the best counterexample.

Trap: Use Latur or Koyna for the peninsular correction.

Q31. How should earthquake content enter the 2026 infrastructure-versus-DRR answer?

A. Claim direct ownership of the whole question B. Discuss only NCS apps C. Use seismic siting, resilient design and lifeline compliance as a bounded dimension D. Replace ecological analysis with zone shares

Answer: C.

Explanation (option-by-option):

- **A.** Primary ownership lies elsewhere.
- **B.** Apps are marginal to the demand.
- **C.** Correct.
- **D.** The question is broader than seismic geography.

Trap: Preserve primary/support PYQ ownership.

Q32. A ministry reports codes, training and a priority list as 'earthquake resilience achieved'. The best assessment is:

A. Accept because three inputs exist B. Reject every input as useless C. Infer casualty reduction D. Treat them as inputs; verify construction, retrofits, continuity and response outcomes

Answer: D.

Explanation (option-by-option):

- **A.** Inputs are not outcomes.
- **B.** Inputs can be useful.
- **C.** Causation is unproven.
- **D.** Correct.

Trap: The safety-outcome firewall prevents success-shaped overclaiming.

PYQS AND ANSWER PRACTICE

DIRECT PYQ - 2021 GS-III Q8 - 10 marks, 150 words

Question: Discuss about the vulnerability of India to earthquake related hazards. Give examples including the salient features of major disasters caused by earthquakes in different parts of India during the last three decades.

Alternative workbook model: India's vulnerability combines wide seismic exposure with unsafe buildings and uneven response capacity. The Himalayan convergence zone, North-East, Andaman and Nicobar Islands and Kachchh face high hazard; about 59 per cent of land lies in Zones III-V. Yet regional zone alone does not determine loss.

Unreinforced masonry, soft-storey buildings, poor connections and vulnerable hospitals and utilities convert shaking into disaster. Latur (1993) showed damaging peninsular risk and rural masonry weakness; Bhuj (2001) exposed severe housing and urban vulnerability; Kashmir (2005) combined weak buildings with mountainous access constraints; Sikkim (2011) affected connectivity and slope stability.

Since precise prediction is unavailable, India must enforce safe new construction, audit and retrofit priority structures, train professionals, use microzonation and land-use control, strengthen lifelines and prepare EOC/IRS response. Earthquake resilience is therefore primarily a compliance and vulnerability-reduction challenge.

2026 bounded application

For GS-III Q7, use earthquake evidence only to support risk-sensitive siting, resilient infrastructure and lifecycle enforcement in ecologically sensitive areas. Primary ownership remains with landslide/GLOF and critical-infrastructure risk.

ORIGINAL MAINS PRACTICE - WORKBOOK MODELS

Original Mains 1 - 10 marks | 150 words

Question: Distinguish earthquake magnitude from intensity and explain the planning value of the distinction.

Model answer: Magnitude estimates energy released at the source and is commonly expressed as moment magnitude for large events. It is broadly one event-level value. Intensity describes observed effects at a location and may be expressed through MMI or MSK scales. It varies with distance, depth, soil amplification, topography and building stock. The distinction matters because magnitude alone cannot predict damage. A moderate event may produce severe local effects on soft soil or weak masonry, while better construction may limit loss under stronger shaking. Intensity observations support rapid damage mapping and response prioritisation; microzonation and site investigation support pre-disaster planning. Thus source magnitude describes the earthquake, whereas intensity connects shaking to local exposure and vulnerability.

Original Mains 2 - 10 marks | 150 words

Question: Why is code adoption insufficient to establish earthquake-safe construction?

Model answer: A code is only the first link in a compliance chain. Safety also requires correct site data, competent structural design, ductile detailing, approved materials, trained masons, faithful workmanship, inspection, as-built verification and maintenance. Approved drawings can be defeated by altered reinforcement or poor connections. VisionIAS therefore lists weak bye-law enforcement, limited professional training and absent licensing systems as connected vulnerabilities. The National Building Code and BIS standards remain essential, but adoption does not prove local enforcement or the quality of an individual building. Governments should combine professional licensing, transparent permits, risk-based inspections, mason training and accountable completion certification. Earthquake safety is achieved through verified implementation, not the existence of a technical document.

Original Mains 3 - 15 marks | 250 words

Question: Explain NDMA's six-pillar framework for seismic safety and its implementation challenges.

Model answer: NDMA's framework covers the full seismic-risk chain. First, earthquake-resistant new construction prevents further accumulation of unsafe stock. Second, selective strengthening and retrofitting addresses existing priority buildings and lifelines. Third, regulation and enforcement translate technical standards into practice. Fourth, awareness and preparedness shape household and institutional action. Fifth, capacity development improves engineers, architects, masons and administrators. Sixth, emergency response organises rescue, medical aid and coordination after impact. Implementation is uneven because building bye-laws are weakly enforced, professional licensing and training are incomplete, construction quality varies and retrofitting requires large finance and temporary relocation. States must identify priority structures, but identification does not prove completed work. Hospitals and EOCs also need power, water, equipment and staff continuity, not only frame safety. The framework succeeds only when all six pillars are treated as one system and progress is measured through verified construction, completed retrofits, drill learning and functional response.

Original Mains 4 - 15 marks | 250 words

Question: Discuss the role and limits of seismic microzonation in urban risk reduction.

Model answer: Seismic microzonation refines a broad regional zone by mapping local soil, basin, slope, groundwater, liquefaction and amplification conditions together with exposure and critical facilities. It helps cities set site-sensitive design requirements, guide land use, locate hospitals and EOCs, preserve access routes and prioritise vulnerable neighbourhoods. It is especially useful where adjacent localities may experience different shaking despite being in the same national zone. Its limits are equally important. Microzonation does not predict the time or magnitude of an earthquake, certify individual buildings or automatically alter development. Maps can become outdated as urbanisation, groundwater and infrastructure change. Risk falls only when planning regulations, building design, permits and retrofitting actually use the findings. Data quality, public access and enforcement capacity therefore matter as much as the map. Microzonation is best understood as a local decision input within a wider compliance system, not as a stand-alone safety outcome.

Original Mains 5 - 20 marks | 250 words

Question: Critically examine India's challenge of retrofitting existing buildings and lifeline infrastructure.

Model answer: Existing stock is the hardest seismic challenge because many buildings predate current standards, engineering records are incomplete and strengthening is costly. A sound programme begins with inventory and rapid screening, followed by detailed audits, priority ranking, retrofit design, finance, execution and quality verification. States are responsible for identifying priority structures, while local bodies support implementation. Hospitals, schools, bridges, dams, EOCs and utility systems merit priority because their failure produces cascading loss. Retrofitting must address both structural performance and functional continuity: a hospital frame that survives but loses power, water or equipment is not resilient. Major constraints include funding, relocation during works, shortage of skilled professionals, procurement quality, incomplete records and political preference for visible new projects. Priority lists or sanctions should not be reported as completed mitigation. India needs phased risk-based programmes, transparent audits, standard retrofit guidance, dedicated finance, third-party verification and continuity testing. New-build enforcement must continue simultaneously, or the vulnerable stock will keep growing.

Original Mains 6 - 20 marks | 250 words

Question: Assess the proposition that earthquake resilience in India is fundamentally a governance and compliance problem.

Model answer: The proposition is persuasive because earthquake occurrence cannot be precisely prevented or predicted. Loss reduction therefore depends on governing exposure and vulnerability. India already has seismic zoning, BIS standards, the National Building Code, NDMA's six pillars, BMTPC technical resources and NCS monitoring. The central deficit is conversion into safe buildings and functioning lifelines. Weak bye-law enforcement, inadequately trained professionals, poor workmanship and incomplete retrofitting break the compliance chain. Governance also shapes land use: dense development on unstable slopes or liquefaction-prone ground can defeat building-level measures. State capacity varies because States identify priority structures and local bodies enforce many controls. Yet governance is not the only issue; scientific mapping, materials, finance and public behaviour remain necessary inputs. The correct verdict is that technology defines possible safety, while governance determines whether it reaches the built environment. India needs transparent permits, trained professionals, verified construction, lifeline audits, retrofit finance and outcome-based monitoring.